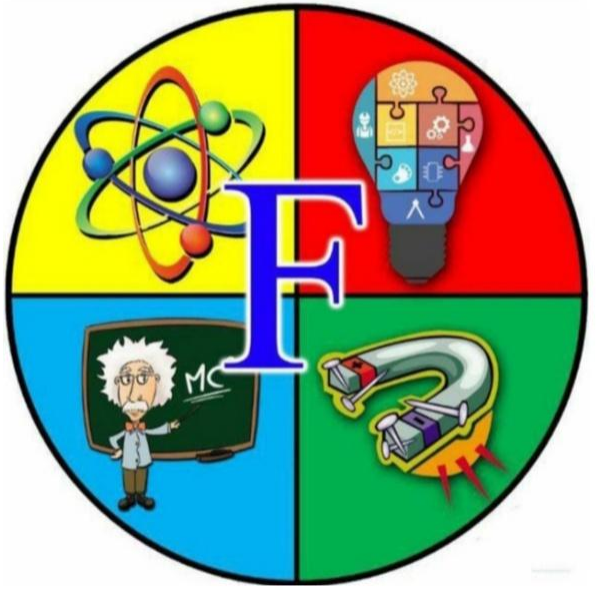
Fizika

9-sinf

Mavzu:

Masalalar yechish



$$n = \frac{c}{\vartheta}$$

$$\frac{n_2}{n_1} = \frac{\vartheta_1}{\vartheta_2}$$

$$n_{21} = \frac{n_2}{n_1} = \frac{\sin \alpha}{\sin \beta}$$

Masala

Yorug‘lik birinchi muhitdan ikkinchi muhitga $45°$ burchak ostida tushib, $30°$ burchak ostida sinib o‘tmoqda.

Birinchi muhitning absolyut nur sindirish ko‘rsatkichi $\sqrt{2}$ ga teng bo‘lsa, yorug‘likning ikkinchi muhitdagi tezligi nimaga teng?

Berilgan:

$c = 3 \cdot 10^8\ m/s$

$\alpha = 45°$

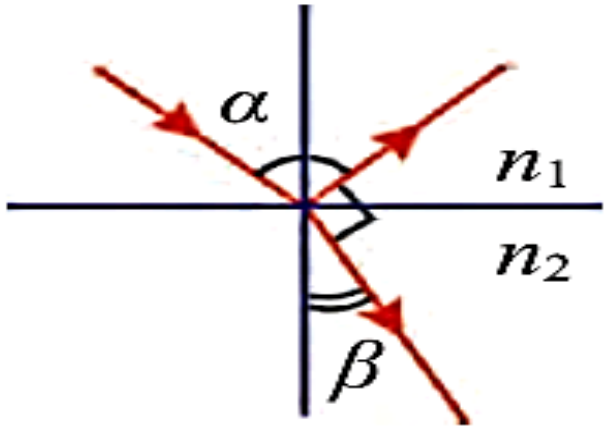
$\beta = 30°$

$n_1 = \sqrt{2}$

Topish kerak: $\vartheta_2 - ?$

Formula:

$$\vartheta_2 = \frac{c}{n_2}$$



$$\frac{n_2}{n_1} = \frac{\sin\alpha}{\sin\beta}$$

$$n_2 = \frac{n_1 \sin\alpha}{\sin\beta} \quad \text{dan} \quad \vartheta_2 = \frac{c}{n_1} \cdot \frac{\sin\beta}{\sin\alpha}$$

Yechish:

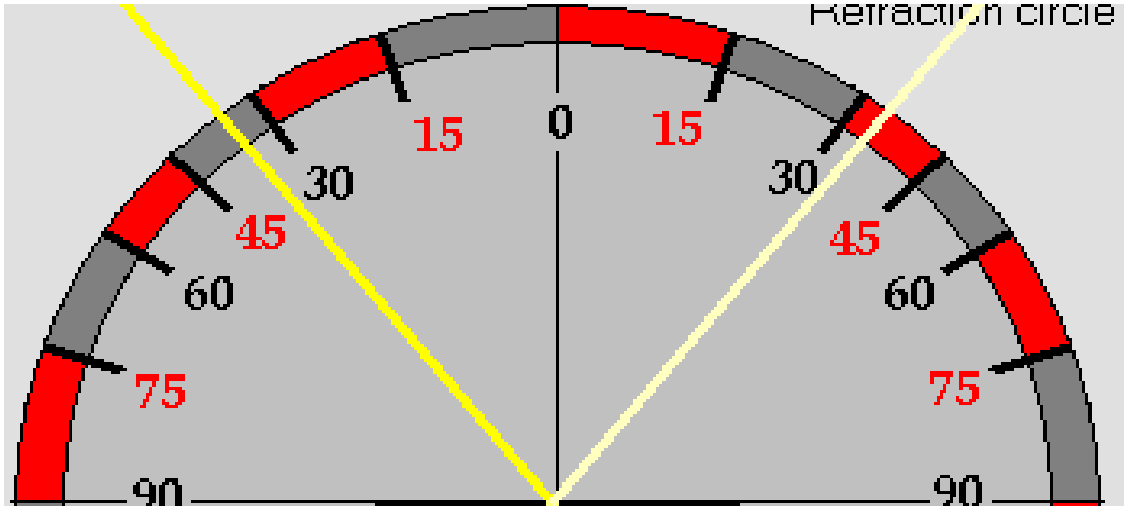
$$\sin 30° = 0{,}5 \qquad \sin 45° = \frac{\sqrt{2}}{2}$$

$$\vartheta_2 = \frac{3 \cdot 10^8\ m/s}{\sqrt{2}} \cdot \frac{0{,}5 \cdot 2}{\sqrt{2}} = 1{,}5 \cdot 10^8\ m/s$$

Javob: $\vartheta_2 = 1,5 \cdot 10^8\ m/s$

25-mashq 1-masala

Tushgan va qaytgan nurlar orasidagi burchak $\mathbf{70°}$ bo‘lishi uchun, nur ko‘zguga qanday burchak ostida tushishi kerak?



$\alpha - ?$

$\alpha + \gamma = 70°$

$\alpha = \gamma$ dan $\alpha = \frac{70°}{2} = 35°$

Javob: **35° burchak ostida tushishi kerak.**

25-mashq 4-masala

Havodan shishaga tushgan va qaytgan nurlar orasidagi burchak $60°$ga teng.

Shishaning sindirish ko‘rsatkichi $1,5$ ga teng bo‘lsa, sinish burchagini toping.

Berilgan:

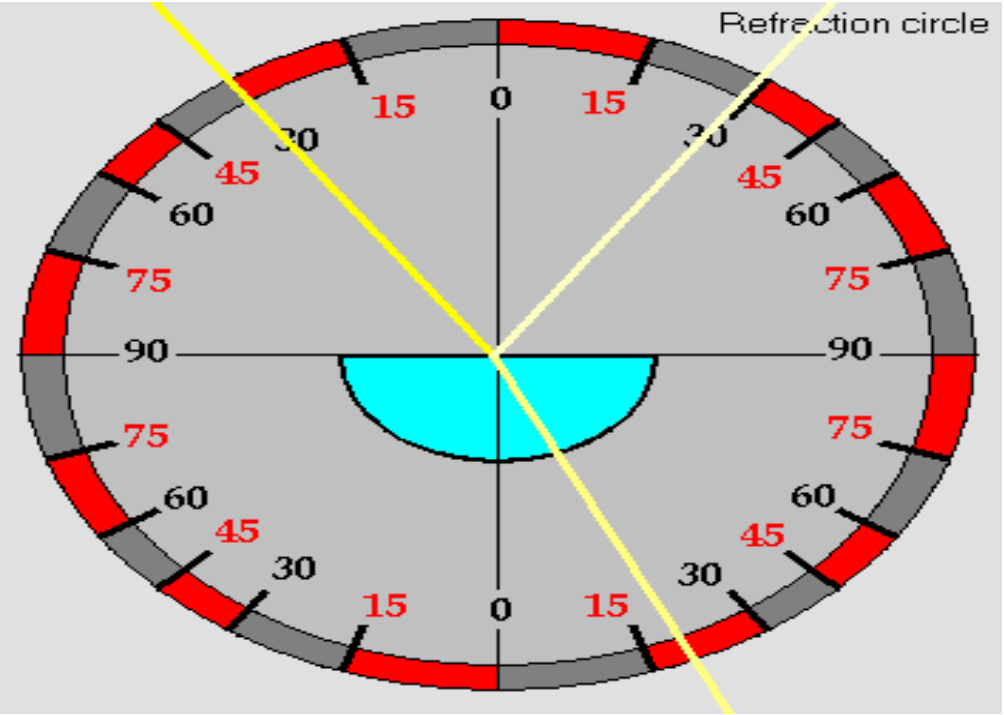
$\varphi = \alpha + \gamma = 60°$

$n_1 = 1$

$n_2 = 1{,}5$

Topish kerak: $\beta - ?$

Formula:



$$\frac{n_2}{n_1} = \frac{\sin\alpha}{\sin\beta}$$

$$\alpha = \gamma \quad \text{dan} \quad \alpha = \frac{\varphi}{2}$$

$$sin\beta = \frac{n_1}{n_2} \cdot \sin\alpha = \frac{n_1}{n_2} \cdot \sin\frac{\varphi}{2}$$

Yechish:

$$\sin 30° = 0{,}5$$

$$sin\beta = \frac{1}{1{,}5} \cdot 0{,}5 \approx 0{,}33$$

$$sin19{,}5° \approx 0{,}33$$

Javob: $\beta \approx 19{,}5°$

25-mashq 5-masala

Nur suvdan shishaga o‘tmoqda. Suvning nur sindirish ko‘rsatkichi **1,33** ga, shishaniki **1,5** ga teng.

Bir xil vaqt ichida shu moddalardan yorug‘lik nuri o‘tgan masofalar nisbati nimaga teng?

Berilgan:

$$n_1 = 1{,}33$$

$$n_2 = 1{,}5$$

$$t_1 = t_2 = t$$

Topish kerak: $\frac{S_1}{S_2} - ?$

Formula:

$$S = \vartheta t$$

$$\frac{S_1}{S_2} = \frac{\vartheta_1 t}{\vartheta_2 t}$$

$$\vartheta_1 = \frac{c}{n_1} \quad \vartheta_2 = \frac{c}{n_2}$$

$$\frac{S_1}{S_2} = \frac{cn_2}{n_1 c} = \frac{n_2}{n_1}$$

Yechish: $\frac{S_1}{S_2} = \frac{1{,}5}{1{,}33} \approx 1{,}13$

Javob: $\frac{S_1}{S_2} \approx 1{,}13$

Mustaqil bajarish uchun topshiriqlar

1. Yassi koʻzguda buyum tasviri koʻzgudan **60 cm** masofada hosil boʻlsa, buyum bilan uning tasviri orasidagi masofa qanday boʻladi?

2. Absolyut sindirish koʻrsatkichi **2** ga teng boʻlgan muhitda yorugʻlik qanday tezlik bilan tarqaladi?